2. The heap tree is complete and left-adjusted. Complete (see figure 2.3) means that if the heap has height `H`, every leaf node is either at level `H` or `H-1`. All the levels are left-adjusted which means that no right sub-tree has a height greater than its left sibling. So, if a leaf is at the same height as an internal node[7], the leaf can't be on the left of that node. Invariants number 1 and 2 are the structural invariants for heaps.
3. Every node holds the highest priority in the subtree rooted at that node.

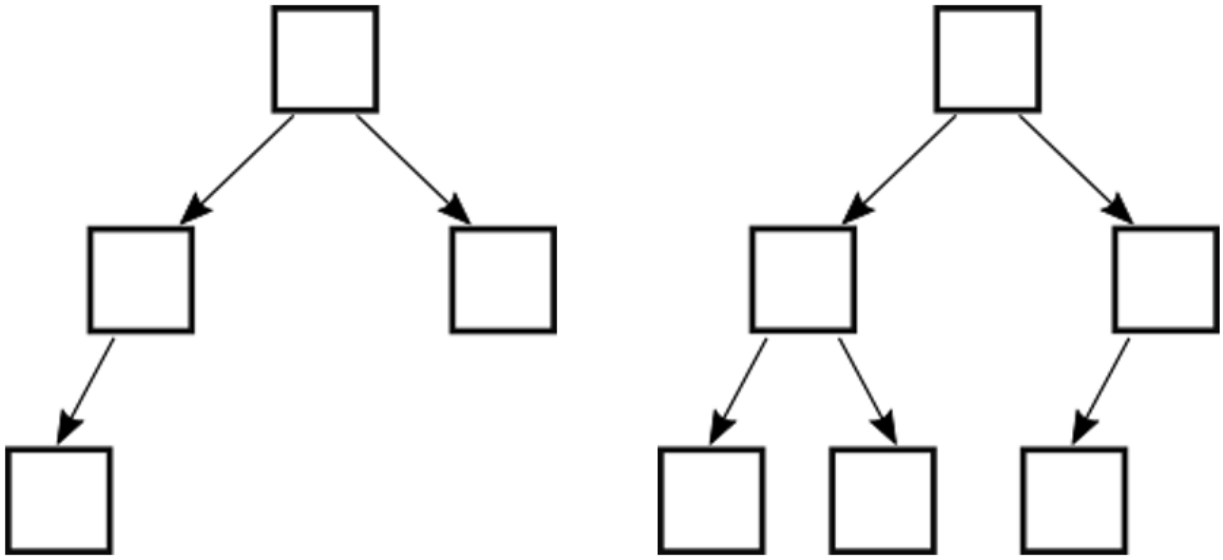


Figure 2.3 Two examples of complete binary trees. All levels in the tree have the maximum possible number of nodes, except (possibly) the last one. All leaves in the last level are left-adjusted: at the previous level, no right subtree has more nodes than its left sibling.

Properties (1) and (2) above allow for the array representation of the heap: supposing we need to store `N` elements, the tree structure is directly and compactly